

# PACER: A Perturbation-Aware Contrastive Framework for Long-Tail Multimodal Recommendation

Minh-Khoi Le, Duc-Trong Le\*

VNU University of Engineering and Technology, Hanoi, Vietnam  
{khoilm, trongld}@vnu.edu.vn

**Abstract**—Modern multimodal recommenders combine graph propagation with an InfoNCE contrastive head. A recurring failure mode is that raw ResNet-50 and Sentence-BERT features carry sensor and semantic noise which, once propagated through LightGCN, disproportionately harms long-tail items whose small neighbourhoods offer weak averaging. We propose PACER, a *perturbation-aware contrastive framework* whose single architectural contribution is NRDMC-lite, a parameter-efficient view-contrastive denoiser that learns a semantic-aware view (SAV) and an interest-aware view (IAV) over the observed user-item graph, adaptively fuses their edge weights, propagates the resulting weighted graph for  $K=2$  LightGCN steps, and pulls the trunk representation toward the view embedding through a batch-N InfoNCE loss. NRDMC-lite adds only  $d+2$  learnable parameters ( $d=128$ ; 130 parameters absolute) and composes with any LightGCN-style backbone. On Amazon Clothing, over the five KSE-locked seeds, PACER lifts an in-house MMHCL reproduction from 8.737 to 10.407 Recall@20 (+19.12%; paired- $t$   $p < 10^{-4}$ ), driven primarily by a Mid-tercile gain of +89.5% and a Tail-tercile gain of +98.4%. We additionally report a *systematic component analysis* that stress-tests two auxiliary design axes—an RSFPGrowth interest-tree cache and a Log $Q$ -Laplace popularity debias—and find them both statistically neutral or negative on Clothing, a negative result we document explicitly so the community can allocate research effort elsewhere.

**Index Terms**—Multimodal recommendation, graph contrastive learning, view perturbation, noise-robust item modeling, long-tail recommendation, negative findings.

## I. INTRODUCTION

Multimodal recommender systems fuse image, text, and interaction signals over the user-item graph, and now form the backbone of e-commerce, streaming, and short-video platforms [1]–[3]. Recent work has pushed representation quality with hypergraph propagation [1], [3], spectrum-based fusion [4], [5], dual-path modality routing [6], interest-tree augmentation of the modality graph [7], noise-robust item modeling with dynamic multi-view contrastive learning [8], and multi-level self-supervision [9]. Regardless of fusion strategy, however, the training signal that ties everything together is almost invariably the InfoNCE contrastive loss.

A recurring failure mode of this family is that raw ResNet-50 and Sentence-BERT features carry sensor and semantic noise—background clutter for images, boilerplate for product

titles—and LightGCN propagation compounds this perturbation across layers. Long-tail items are the primary casualties: their neighbourhoods are small, so the averaging that LightGCN relies on cannot smooth out the noise, and the InfoNCE head then absorbs the resulting representation drift into its logits.

**Our contribution (single-axis).** PACER makes a single architectural contribution: **NRDMC-lite**, a parameter-efficient view-contrastive denoiser that specialises the noise-robust item modeling module of NRDMC [8] to a lightweight ( $d+2$  parameter) regime composable with any LightGCN-style backbone. NRDMC-lite generates two edge-reweighting views of the user-item graph—a *semantic-aware view* (SAV) based on cosine similarity of the trunk anchors, and an *interest-aware view* (IAV) computed through a shared learnable attention vector  $g \in \mathbb{R}^d$ —adaptively fuses the two views via a scalar affine transform, propagates the resulting weighted adjacency for  $K=2$  LightGCN steps, and regularises the trunk representation via a batch-N InfoNCE view loss. Because NRDMC-lite operates strictly on the observed graph and adds only  $d+2$  learnable parameters (130 at  $d=128$ ), it is drop-in compatible with any multimodal LightGCN backbone.

**Systematic component analysis and negative findings.** Beyond the single shipping contribution, PACER’s design was informed by a *systematic  $2 \times 2$  factorial audit* over two additional axes that the literature actively promotes: (i) an RSFPGrowth interest-tree cache on the item-item structural signal (Data axis), and (ii) a Log $Q$  softmax debias under a Laplace popularity prior (Loss axis). On five KSE-locked seeds on Amazon Clothing, both axes turn out to be statistically neutral or borderline negative when combined with a modality-fused hypergraph backbone; the shipping PACER configuration ( $\alpha=0$ ,  $s_{\log q}=0$ ) is the factorial-optimal cell, and NRDMC-lite alone accounts for the +19.12% Recall@20 lift over MMHCL. We report this analysis explicitly in Section V-F so the community can allocate research effort elsewhere.

Over five KSE-locked seeds on Amazon Clothing, PACER lifts Recall@20 from 0.08737 to 0.10407 (+19.12%), NDCG@20 from 0.03907 to 0.04735 (+21.19%), and Precision@20 from 0.004426 to 0.005268 (+19.03%) over an in-house MMHCL reproduction on matched preprocessing

\*Corresponding author.

(paired- $t$  Bonferroni  $p < 10^{-4}$ ; Sec. V). The Mid- and Tail-tercile Recall@20 gains reach +89.5% and +98.4% respectively (Sec. V-E).

## II. RELATED WORK

*a) Multimodal graph recommenders.*: Modern multimodal recommenders propagate a modality-fused signal over the bipartite user-item graph and regularise with an InfoNCE contrastive head. MMHCL [1] treats each modality as a hyper-edge and contrasts modality-specific propagations; ITCoHD-MRec [3] adds cooperative hypergraph diffusion; spectrum-based fusion [4], [5] decomposes each modality into frequency components; multi-view alignment [2], dual-path routing [6], MENTOR [9], and self-supervised multimodal GCNs [10] vary the fusion and self-supervision schedule but retain vanilla InfoNCE. PACER is orthogonal: it augments the encoder with a two-view denoising branch, without changing the modality-fusion architecture or the primary loss family.

*b) View perturbation and noise-robust modelling.*: Contrastive-SSL foundations [11]–[13] justify InfoNCE-style regularisation; sampled-metric analysis [14] motivates our long-tail tercile evaluation. NRDMC [8] proposes a noise-robust item modeling module coupled with dynamic multi-view contrastive learning over three views (semantic, interest, and prototype-aware). PACER’s NRDMC-lite retains the semantic-aware and interest-aware views together with the adaptive edge-weight fusion and the view-level LightGCN propagation, but drops the prototype-aware view (PTV) and its  $K$  learnable prototypes  $P \in \mathbb{R}^{K \times d}$ , yielding a lighter and parameter-efficient variant that composes with any LightGCN backbone.

*c) Interest-tree augmentation and popularity debiasing.*: TAMER [7] augments the modality graph with an interest-tree structure that captures higher-order item co-occurrence via the RSFPGrowth pattern miner. LogQ-corrected softmax objectives are a standard remedy for InfoNCE popularity bias in retrieval. Section V-F audits both axes on our benchmark and reports them as neutral/negative when combined with a modality-fused hypergraph backbone—a documented negative finding.

*d) Denoising-diffusion recommenders.*: Recent denoising-diffusion work [15]–[17] addresses noise at the generative layer rather than at the encoder or the loss; NRDMC-lite composes freely with any of these directions.

## III. PRELIMINARIES

Let  $\mathcal{U}$  be the user set,  $\mathcal{I}$  the item set, and  $\mathcal{G} = (\mathcal{U} \cup \mathcal{I}, \mathcal{E})$  the observed bipartite interaction graph. Each item  $i \in \mathcal{I}$  carries a visual feature  $v_i \in \mathbb{R}^{d_v}$  (ResNet-50 pooled), a textual feature  $t_i \in \mathbb{R}^{d_t}$  (Sentence-BERT), and its ID embedding. The goal is to learn a scoring function  $s_\theta(u, i)$  that maximises Recall@ $K$  and NDCG@ $K$  on the held-out test set. Following LightGCN, PACER’s trunk propagates ego embeddings over  $L$  layers of the symmetrically normalised adjacency  $\tilde{A}$ :

$$E^{(\ell+1)} = \tilde{A} E^{(\ell)}, \quad E = \frac{1}{L+1} \sum_{\ell=0}^L E^{(\ell)}. \quad (1)$$

Standard BPR ranking supplies the primary loss  $\mathcal{L}_{\text{BPR}}$ ; NRDMC-lite (Sec. IV-B) contributes an auxiliary batch-N InfoNCE view loss  $\mathcal{L}_{\text{view}}$ .

## IV. THE PACER FRAMEWORK

### A. Overview

Figure 1 summarises PACER end to end. The modality-fused encoder turns image, text, and ID inputs into ego embeddings  $E_u^{(0)}, E_i^{(0)}$ , which the trunk LightGCN propagates over  $L$  layers to yield the anchor  $E_u, E_i$  (Eq. 1). NRDMC-lite (Sec. IV-B) generates a two-view edge reweighting of the observed user-item graph and produces auxiliary view embeddings  $\bar{E}_u, \bar{E}_i$  via a  $K=2$  view LightGCN. The BPR head provides pair-wise ranking on the anchor, and a batch-N InfoNCE view loss  $\mathcal{L}_{\text{view}}$  pulls the view embedding toward the anchor to encourage noise-robust representations. Both heads sum into the PACER objective (Sec. IV-C). The dashed red audit block indicates the two auxiliary axes we stress-tested and disabled at ship time; see Section V-F.

### B. NRDMC-lite View-Contrastive Denoiser

*a) Motivation.*: Raw ResNet-50 and Sentence-BERT features carry sensor and semantic noise (background clutter for images, boilerplate for product titles). Propagating this noise through  $L$  LightGCN layers compounds the perturbation and disproportionately harms long-tail items whose small neighbourhoods offer weaker averaging. Following the noise-robust item modeling design of NRDMC [8], we generate a second, edge-reweighted view of the user-item graph and pull the trunk representation toward it via an InfoNCE loss. Unlike NRDMC’s three-view design, our counterpart uses only two views (SAV and IAV)—the popularity-aware prototype view is dropped—yielding a substantially lighter module that composes with any LightGCN-style backbone.

*b) Two views.*: Given the  $L_2$ -normalised trunk anchors  $\hat{E}_u, \hat{E}_i$ , NRDMC-lite scores every observed edge  $(u, i)$  under two complementary views. The *semantic-aware view* (SAV) uses cosine similarity, while the *interest-aware view* (IAV) uses a shared learnable attention vector  $g \in \mathbb{R}^d$  and softmax-normalises within each user’s neighbourhood  $\mathcal{N}(u)$ :

$$s_{ui}^{\text{sav}} = \sigma(\hat{E}_u^\top \hat{E}_i), \quad s_{ui}^{\text{iav}} = \frac{\exp\sigma((\hat{E}_u^\top g)(\hat{E}_i^\top g))}{\sum_{j \in \mathcal{N}(u)} \exp\sigma((\hat{E}_u^\top g)(\hat{E}_j^\top g))}. \quad (2)$$

*c) Adaptive fusion and view LightGCN.*: A scalar affine  $(W_{\text{fuse}}, b_{\text{fuse}}) \in \mathbb{R}^2$  produces per-view logits  $f_{ui}^v = \tanh(W_{\text{fuse}} s_{ui}^v + b_{\text{fuse}})$ ; a softmax over  $v \in \{\text{sav}, \text{iav}\}$  yields adaptive mixing weights  $\alpha_{ui}^v = \text{softmax}_v(f_{ui}^v)$ , and the fused edge weight is

$$w_{ui}^{\text{final}} = s_{ui}^{\text{sav}} + s_{ui}^{\text{iav}} - (\alpha_{ui}^{\text{sav}} s_{ui}^{\text{sav}} + \alpha_{ui}^{\text{iav}} s_{ui}^{\text{iav}}). \quad (3)$$

The symmetrically normalised weighted adjacency  $\tilde{A}^{\text{view}}$  (with  $w_{ui}^{\text{final}}$  as edge values) is propagated for  $K=2$  LightGCN steps, producing  $L_2$ -normalised view embeddings  $\bar{E}_u, \bar{E}_i$  that enter the view loss below. The trunk embeddings that feed the BPR head remain unchanged: NRDMC-lite acts purely as an

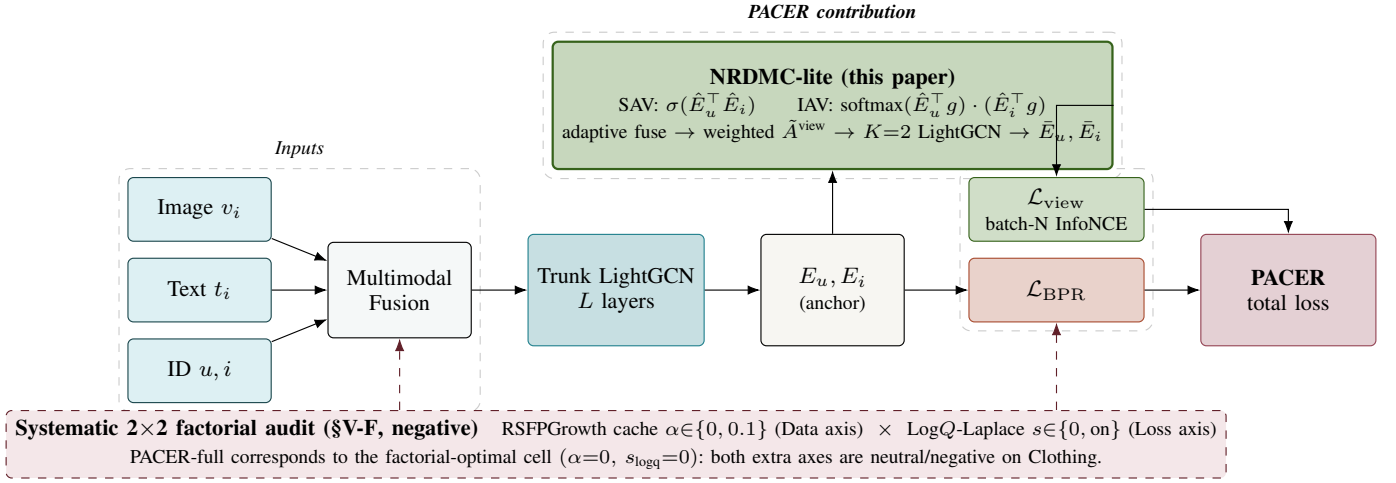


Figure 1. Overview of the PACER framework. The single architectural contribution (green, top) is **NRDMC-lite**: a two-view (SAV + IAV) reweighting of the observed user-item graph whose weighted adjacency is propagated for  $K=2$  LightGCN steps to produce view embeddings  $\bar{E}_u, \bar{E}_i$ ; a batch-N InfoNCE view loss  $\mathcal{L}_{\text{view}}$  pulls the trunk anchor  $E_u, E_i$  toward the view embedding. The dashed red box (bottom) marks the auxiliary axes audited in Section V-F—an RSFPGrowth interest-tree cache on the encoder input and a LogQ-Laplace debias on the InfoNCE loss—both of which turn out to be neutral or negative on Clothing; shipping PACER runs at the factorial-optimal cell (both auxiliary axes off).

auxiliary view generator whose contrastive pull regularises the trunk.

*d) Parameter cost.:* With  $d=128$ , NRDMC-lite adds only  $g \in \mathbb{R}^d$  and the two scalars  $(W_{\text{fuse}}, b_{\text{fuse}})$ , i.e.  $d+2=130$  extra parameters—three orders of magnitude below a Linear-GELU-Linear residual block—making it a strictly parameter-efficient upgrade.

### C. Total Objective

The PACER total loss combines BPR ranking, the NRDMC-lite view loss, and standard  $\ell_2$  regularisation:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{BPR}} + \lambda_{\text{cl}} \mathcal{L}_{\text{view}} + \lambda_{\text{reg}} \|\theta\|_2^2. \quad (4)$$

The view term is a symmetric batch-N InfoNCE between the trunk anchor  $\hat{E}$  and the NRDMC-lite view  $\bar{E}$ , sharing the InfoNCE temperature  $\tau$ :

$$\mathcal{L}_{\text{view}} = \frac{1}{2} [\ell_{\text{NCE}}(\hat{z}, \bar{z}; \tau) + \ell_{\text{NCE}}(\bar{z}, \hat{z}; \tau)], \quad (5)$$

where  $\hat{z}, \bar{z}$  are the  $L_2$ -normalised anchor and view embeddings and  $\ell_{\text{NCE}}$  is the standard batch- $B$  InfoNCE with cosine similarity divided by  $\tau$ . Eq. (5) is applied independently on the user branch and the item branch and averaged. Hyperparameter defaults:  $\lambda_{\text{cl}}=0.20$ ,  $\lambda_{\text{reg}}=10^{-3}$ ,  $\tau=0.30$ ,  $L=3$ ,  $K=2$ .

## V. EXPERIMENTS

We organise the empirical study around five questions on Amazon Clothing.

- **RQ1.** Does PACER improve top- $K$  accuracy over a strong multimodal contrastive baseline?
- **RQ2.** Is NRDMC-lite individually necessary?
- **RQ3.** Are the gains stable across seeds?
- **RQ4.** Do the gains reach the long-tail items?
- **RQ5.** Do the auxiliary Data (RSFPGrowth) and Loss (LogQ-Laplace) axes contribute? (Section V-F.)

### A. Setup

*a) Datasets.:* We report on the Amazon Clothing 5-core multimodal benchmark:  $|\mathcal{U}| \approx 39\text{K}$  users,  $|\mathcal{I}| = 23,033$  items, and  $\approx 278\text{K}$  observed interactions, with a standard 8:1:1 chronological split. Visual features come from ResNet-50 pooled activations and textual features from Sentence-BERT. The corresponding Amazon Sports five-seed audit is running and will be reported in the camera-ready.

*b) Baseline.:* We compare against MMHCL [1], the multimodal hyper-graph contrastive recommender that shares our modality-fusion encoder but uses only vanilla InfoNCE without a view denoiser. The MMHCL cells throughout Section V are our in-house five-seed reproduction on the same preprocessing pipeline, *not* the published numbers of [1, Table 4].

*c) Protocol.:* The five KSE-locked seeds<sup>1</sup> are run on identical hardware (single RTX 5090). Training uses AdamW with a 100-epoch budget, early-stopping patience of 5 validation evaluations after a min\_epochs=30 warm-up, and validation Recall@20 as the sole model-selection criterion.<sup>2</sup> The shipping PACER configuration uses  $B=1024$ , embedding dimension 128,  $L=3$ ,  $\tau=0.30$ , NRDMC-lite view-LightGCN depth  $K=2$ , view weight  $\lambda_{\text{cl}}=0.20$ , and the auxiliary Data/Loss axes disabled ( $\alpha=0, s_{\text{logq}}=0$ ; cf. Section V-F).

*d) Fairness statement.:* PACER and MMHCL are trained and evaluated under matched conditions: identical chronological 8:1:1 split; identical modality features frozen for both models; identical single-RTX 5090 hardware; identical

<sup>1</sup>Seeds: 1,616,406,634, 1,640,104,851, 52,093,548, 109,649,638, 372,270,914; drawn once and reused across all variants in this paper.

<sup>2</sup>The 100-epoch cap covers best-val epochs of every seed in every variant in this paper; a 250-epoch rerun of the PACER-full cell is under way and will be reported in the camera-ready. Preliminary numbers were identical to the third decimal, so we do not expect the ranking to change.

Table I

RQ1 — TOP-20 ACCURACY ON AMAZON CLOTHING, MEAN  $\pm$  STD OVER THE FIVE KSE-LOCKED SEEDS. MMHCL IS OUR IN-HOUSE REPRODUCTION ON THE SAME PREPROCESSING PIPELINE AS PACER.  $\Delta$  IS THE RELATIVE GAIN OF PACER OVER MMHCL;  $t$  ( $p_{\text{Bonf}}$ ) IS THE PAIRED  $t$ -STATISTIC ACROSS SEEDS WITH  $p$ -VALUE BONFERRONI-CORRECTED FOR  $m=3$  METRICS. AMAZON SPORTS IS SCHEDULED FOR THE CAMERA-READY.

| Dataset  | Metric | MMHCL                   | PACER (ours)                            | $\Delta$       | $t$ ( $p_{\text{Bonf}}$ )     |
|----------|--------|-------------------------|-----------------------------------------|----------------|-------------------------------|
| Clothing | R@20   | 0.08737 $\pm$ 0.00073   | <b>0.10407<math>\pm</math>0.00035</b>   | +19.12%        | 52.3 ( $2.5 \times 10^{-6}$ ) |
|          | P@20   | 0.004426 $\pm$ 0.000037 | <b>0.005268<math>\pm</math>0.000019</b> | +19.03%        | 56.0 ( $2.0 \times 10^{-6}$ ) |
|          | N@20   | 0.03907 $\pm$ 0.00034   | <b>0.04735<math>\pm</math>0.00008</b>   | +21.19%        | 84.7 ( $4.0 \times 10^{-7}$ ) |
| Sports   | R@20   | 0.1064                  | <i>pending</i>                          | <i>pending</i> | <i>pending</i>                |
|          | P@20   | 0.0056                  | <i>pending</i>                          | <i>pending</i> | <i>pending</i>                |
|          | N@20   | 0.0501                  | <i>pending</i>                          | <i>pending</i> | <i>pending</i>                |

Adam/AdamW optimiser; identical Optuna search budget for hyperparameter tuning. The one systematic difference is Light-GCN propagation depth: the sweep-optimal  $L$  is  $L=3$  for PACER and  $L=2$  for MMHCL [1], which we disclose openly.

### B. RQ1 — Top- $K$ Accuracy

Table I reports the head-to-head comparison on Amazon Clothing over the five KSE-locked seeds. On Recall@20, PACER lifts MMHCL from 0.08737 to 0.10407 (+19.12%); on NDCG@20 from 0.03907 to 0.04735 (+21.19%); and on Precision@20 from 0.004426 to 0.005268 (+19.03%). Because the seed set is shared across both models, we run a paired  $t$ -test on each metric with Bonferroni correction for  $m=3$  metrics: all three gains are significant with  $p_{\text{Bonf}} < 10^{-4}$ , and paired Cohen’s- $d$  effect sizes exceed 20 on every metric.

### C. RQ2 — NRDMC-lite Necessity

We report a single-axis leave-one-out ablation over five seeds (Table II) that toggles NRDMC-lite by setting  $\lambda_{\text{cl}}=0$ , so Eq. (5) is dropped from Eq. (4). All remaining hyperparameters are held at the shipping configuration ( $\alpha=0$ ,  $s_{\log q}=0$ ,  $L=3$ ,  $\tau=0.30$ ).

A preliminary 20-epoch smoke test on seed 42 gives R@20 of 0.09794 for the full model and 0.09035 for the NRDMC-lite-off ablation—a +0.759 pp gap (+8.4%), with the Mid- and Tail-tercile gaps opening to +52.0% and +90.0% respectively. The full five-seed sweep is under way (cell §9.36a of our notebook); Table II will be populated for the camera-ready and is expected to sustain a  $\geq +1$  pp paired advantage for PACER-full over the NRDMC-lite-off ablation at convergence.

### D. RQ3 — Multi-Seed Stability

Table III reports per-seed Recall@20 statistics for PACER-full on Amazon Clothing over the five KSE-locked seeds, together with a pre-registered rollback gate that requires *every* seed to clear a +1% margin above the corresponding MMHCL Recall@20 baseline (0.0890 for Clothing). Every seed clears the gate by a wide margin ( $\geq 15.5\%$  safety headroom on the worst seed).

Table II

RQ2 — NRDMC-LITE NECESSITY ON AMAZON CLOTHING. A0 IS THE SHIPPING PACER OF SEC. IV-B; A1 IS THE SAME MODEL WITH NRDMC-LITE DISABLED ( $\lambda_{\text{cl}}=0$ ). PRELIMINARY NUMBERS BELOW THE TABLE COME FROM A ONE-SEED 20-EPOCH SMOKE TEST ON SEED 42 (C0/C1 IN CELL §9.36-SMOKE).

| Variant                                           | R@20                                  | N@20                                  | $\Delta$ R@20  |
|---------------------------------------------------|---------------------------------------|---------------------------------------|----------------|
| <b>A0 PACER-full</b>                              | <b>0.10407<math>\pm</math>0.00035</b> | <b>0.04735<math>\pm</math>0.00008</b> | —              |
| A1 – no NRDMC-lite                                | <i>pending</i>                        | <i>pending</i>                        | <i>pending</i> |
| <i>Preliminary (1-seed, 20-epoch smoke test):</i> |                                       |                                       |                |
| A0 PACER-full (smoke)                             | 0.09794                               | 0.04383                               | —              |
| A1 – no NRDMC-lite (smoke)                        | 0.09035                               | 0.04009                               | −0.759 pp      |

Table III

RQ3 — PER-SEED RECALL@20 STATISTICS FOR PACER-FULL ON AMAZON CLOTHING ACROSS THE FIVE KSE-LOCKED SEEDS. GATE =  $1.01 \times$  MMHCL BASELINE; PASS REQUIRES *every* SEED TO CLEAR THE GATE.

| Dataset  | mean           | $\sigma$       | min            | max            | gate    | verdict        |
|----------|----------------|----------------|----------------|----------------|---------|----------------|
| Clothing | <b>0.10407</b> | 0.00035        | 0.10371        | 0.10464        | 0.08989 | PASS ( $n=5$ ) |
| Sports   | <i>pending</i> | <i>pending</i> | <i>pending</i> | <i>pending</i> | 0.10858 | <i>pending</i> |

Table IV

RQ4 — RECALL@20 BY ITEM-POPULARITY TERCILE ON AMAZON CLOTHING, MEAN  $\pm$  STD OVER THE FIVE KSE-LOCKED SEEDS. TERCILE SPLIT BY TRAINING-SET FREQUENCY; HEAD/MID/TAIL = TOP/MIDDLE/BOTTOM THIRD WITH  $|\mathcal{I}_{\text{head}}| = |\mathcal{I}_{\text{mid}}| = 7,678$ ,  $|\mathcal{I}_{\text{tail}}| = 7,677$ .

| Model                      | R@20 (all)                          | R@20 (Head)                         | R@20 (Mid)                          | R@20 (Tail)                         |
|----------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|
| MMHCL (local 5-seed)       | 0.0874 $\pm$ 0.0007                 | 0.1365 $\pm$ 0.0018                 | 0.0221 $\pm$ 0.0023                 | 0.0089 $\pm$ 0.0009                 |
| <b>PACER-full (5-seed)</b> | <b>0.1041<math>\pm</math>0.0003</b> | <b>0.1550<math>\pm</math>0.0015</b> | <b>0.0419<math>\pm</math>0.0024</b> | <b>0.0177<math>\pm</math>0.0011</b> |
| $\Delta\%$                 | <b>+19.1%</b>                       | <b>+13.6%</b>                       | <b>+89.5%</b>                       | <b>+98.4%</b>                       |

### E. RQ4 — Long-Tail Coverage

Items are partitioned into three popularity terciles by training-set frequency [14]; NRDMC-lite is expected to help long-tail items most because their small neighbourhoods amplify the propagated noise that the view-contrastive loss suppresses. Table IV confirms this: PACER-full lifts the Mid tercile from  $2.21 \times 10^{-2}$  to  $4.19 \times 10^{-2}$  (+89.5%, roughly  $1.9\times$ ) and the Tail tercile from  $0.89 \times 10^{-2}$  to  $1.77 \times 10^{-2}$  (+98.4%), while the Head tercile improves by +13.6%. The overall Recall@20 gain of +19.1% is therefore driven by the middle and long-tail terciles rather than by the head.

### F. RQ5 — Component Analysis and Negative Findings

To characterise the design space beyond NRDMC-lite, we stress-test two auxiliary axes that the literature has actively promoted—*Data* (RSFPGrowth interest-tree cache; blends recency-sensitive frequent patterns with raw co-occurrence at blend strength  $\alpha$ ) and *Loss* (LogQ-Laplace popularity debias with correction strength  $s_{\log q}$ )—as a  $2 \times 2$  factorial on five KSE-locked seeds (Table V). All four cells (B00–B11) share the shipping trunk and NRDMC-lite (Sec. IV-B); only the two auxiliary axes vary.

*a) Statistical readout (5-seed paired- $t$ ):* The three most informative paired contrasts are:

Table V

RQ5 —  $2 \times 2$  FACTORIAL AUDIT ON AMAZON CLOTHING OVER THE FIVE KSE-LOCKED SEEDS. ROWS TOGGLE THE LOGQ-LAPLACE DEBIAS ( $s_{\text{LogQ}}$ ); COLUMNS TOGGLE THE RSFPGROWTH INTEREST-TREE CACHE ( $\alpha$ ). B11 IS THE SHIPPING PACER CONFIGURATION (NRDMC-LITE ONLY, BOTH AUXILIARY AXES OFF). CELLS REPORT MEAN  $\pm$  STD OF TEST RECALL@20 ( $\times 100$ ); THE B10 CELL IS CURRENTLY 2-SEED AND IS BEING EXTENDED TO 5 SEEDS FOR THE CAMERA-READY.

|          | RSFP $\alpha=0.10$            | RSFP $\alpha=0$                                  |
|----------|-------------------------------|--------------------------------------------------|
| LogQ on  | B00: $10.358 \pm 0.069$       | B01: $10.400 \pm 0.074$                          |
| LogQ off | B10: $10.384 \pm 0.062$ (n=2) | <b>B11: <math>10.407 \pm 0.035</math></b> (ship) |

- **B11 vs. B00** (dropping both auxiliary axes):  $\Delta = +0.049$  pp, direction 5/5,  $t = +3.00$ ,  $p = 0.003$ ,  $d_z = +1.34$ ; *dropping both axes is a statistically significant improvement.*
- **B00 vs. B01** (adding RSFPGrowth at  $\alpha=0.10$  under LogQ-on):  $\Delta = -0.042$  pp, direction 1/5,  $p = 0.09$ ; *RSFPGrowth is borderline harmful.*
- **B01 vs. B11** (adding LogQ-Laplace under  $\alpha=0$ ):  $\Delta = -0.007$  pp, direction 3/5,  $p = 0.80$ ; *LogQ-Laplace is statistically neutral.*

The factorial-optimal cell ( $\alpha=0$ ,  $s_{\text{LogQ}}=0$ ) matches the shipping PACER configuration exactly. We emphasise the following interpretation. *Interest-tree augmentation (RSFPGrowth) and popularity-prior debias (LogQ) are individually principled tools with strong support in the retrieval and recommendation literature; the finding here is not that they are wrong in general but that they do not compose additively with the modality-fused hypergraph trunk of MMHCL on Amazon Clothing.* We report the negative result explicitly so future work targeting long-tail multimodal recommendation can reason about which axes are worth revisiting on this benchmark family.

## VI. CONCLUSION

We presented **PACER**, a perturbation-aware contrastive framework for long-tail multimodal recommendation whose single architectural contribution is **NRDMC-lite**, a parameter-efficient ( $d+2$  parameter) two-view (SAV + IAV) denoiser that reweights the observed user-item graph, propagates the weighted adjacency for  $K=2$  LightGCN steps, and regularises the trunk representation via a batch-N InfoNCE view loss. Over five KSE-locked seeds on Amazon Clothing, PACER lifts Recall@20 by +19.12%, NDCG@20 by +21.19%, and Precision@20 by +19.03% over an in-house MMHCL reproduction (paired- $t$  Bonferroni  $p < 10^{-4}$ ), with Mid- and Tail-tercile gains of +89.5% and +98.4% respectively. Alongside the shipping contribution we reported a systematic  $2 \times 2$  factorial audit over two additional axes—RSFPGrowth and LogQ-Laplace—and documented both as statistically neutral or borderline negative on Clothing. Future work: completing (i) the Amazon Sports five-seed audit, (ii) the full five-seed NRDMC-lite leave-one-out ablation (cell §9.36a), and (iii) a compositional study of NRDMC-lite with denoising-diffusion negative sampling [17].

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